

**Towards a Multi-Modal and City-Scale Approach for Identification and
Detection of Cyberbullying in Social Networks within Smart Cities: Project
Proposal**

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Abstract

Cyberbullying can spread quickly across the digital channels used in smart cities, such as community forums, service-app feedback threads, and public social media. Text-only detectors have improved, but real moderation still faces three common issues. First, bullying cues often depend on who is talking to whom and how a thread develops. Second, harmful content is rare, so the data is strongly imbalanced. Third, many posts mix text with images and short videos. In this project, we build a practical multi-modal pipeline that combines text, thread and user-session context, Instagram images, and Vine videos represented by a small set of sampled keyframes. We compare classic baselines such as TF-IDF with linear models and transformer baselines such as BERT and RoBERTa, then evaluate simple fusion models that combine text and context embeddings with vision embeddings extracted by a pretrained model such as CLIP. To stay within a realistic course compute budget, we keep large encoders frozen, cache embeddings, and train only small fusion and classification layers. We report macro-F1 and AUPRC, plus ablations by modality, calibration and operating-point analysis, and a short error study that matches moderation use.

Keywords: cyberbullying detection, smart cities, multi-modal learning, transformers, reproducibility, robustness

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Introduction and Motivation

Smart-city infrastructure increasingly relies on digital platforms such as community reporting apps, public service feedback channels, and city-level discussion forums to support resident participation and civic engagement. When residents report neighbourhood issues, comment on public services, or engage in everyday communication through these channels, the same open platforms that support civic life can also become spaces where targeted harassment and cyberbullying occur. If this is not addressed, such content can discourage participation and reduce the public trust that smart-city systems are designed to build.

Automated detection is a practical first step: not to replace human judgement, but to help moderation teams by flagging potentially harmful content for review before it spreads. In practice, a city-scale platform must serve at least three distinct groups: **citizens or victims** who need to assess and document potential harassment directed at them; **content posters** who may benefit from a pre-submission self-check to avoid unintentional harm; and **platform moderators** who triage flagged content and make final decisions with institutional accountability. This human-in-the-loop structure means the classifier’s role is not to issue verdicts, but to provide risk scores and supporting evidence that a human reviewer can act on. In this context, overall accuracy is not the most useful measure—a classifier that gets 90% accuracy simply by labelling everything as benign is not useful for moderation at all. What matters more is whether the system can reliably identify minority harassment types, stay consistent across different community groups on a city-scale platform, and give confidence scores that are meaningful enough for setting moderation thresholds.

These concerns point to three specific challenges that motivate this work [5]. User-generated text in civic platforms tends to be informal and community-specific: slang,

local references, and platform conventions can vary a lot across districts, meaning a model trained on one community’s data may perform much worse on another’s. Harmful content is also relatively rare, which means standard training objectives tend to under-represent minority harassment types in both training and evaluation. Finally, the text itself does not always make intent clear—the same phrase can be harmless in one thread and targeted abuse in another, depending on who is involved and the context around it. This last point suggests that adding non-textual signals—such as timing, interaction structure, and user-level metadata—may improve detection where text alone is not enough [13].

Problem Statement

We aim to build a city-scale cyberbullying detection workflow that is both **accountable** and **context-aware**. We use consistent evaluation across model families, and we incorporate signals beyond raw text. We prototype this workflow as the **CitySafe** pilot application. To satisfy the privacy-first constraint, CitySafe includes a **Local Evidence Kit** that stores analysis records on the user device by default, and requires explicit consent before any content is uploaded or shared with moderators.

Two governance principles constrain the design from the outset. First, **human-in-the-loop**: classifier outputs are assistive evidence, such as risk scores, supporting context, and per-class confidence, rather than automated verdicts. Final decisions remain with human reviewers, so the model should produce confidence scores that are well-calibrated, not only binary labels. Second, **privacy-first**: a deployable system should minimize data retention, support explicit user consent before any content is shared externally, and operate on the minimum information needed for each use case, for example, coarse region metadata instead of precise location. These constraints shape our evaluation priorities. Calibration and threshold selection are first-class metrics, not afterthoughts. We focus on moderation-oriented operating points, meaning the trade-off between recall and precision under class imbalance, rather than aggregate accuracy.

In this proposal, we use the term “multi-modal” in a broad and practical way. It

means we combine different kinds of evidence that a moderation system can realistically access. Text is always present, and we also use thread or user-session context. When media is linked to a post, we add frozen visual features from images and from a few sampled video frames. Audio is treated as optional and is only used if we can reliably extract a sound track from the video files. This is closer to feature-augmented text classification than full end-to-end multi-modal modelling, but it fits our dataset and compute budget and still lets us test whether extra signals help in practice [13].

Specifically, all models will use text as input, and our proposed system additionally uses **contextual signals** extracted from the dataset:

- **Structured context signals:** post-level metadata such as timestamp bins, platform or channel identifiers, and simple interaction statistics such as reply count and mention density, when the dataset provides them.
- **Extended signals:** cross-platform linkage identifiers such as URL or media IDs, plus simple conversation-graph statistics derived from reply or mention edges.

This framing reflects how harassment operates in practice: the same words carry different risk depending on context, timing, and the interaction history surrounding them.

Objectives and Research Questions

Objectives

1. Build a reproducible data-to-metrics pipeline for cyberbullying detection that fits smart-city moderation. The workflow is privacy-first and supports human review.
2. Set up strong baselines, both classic ML and transformer text models, using the same splits, a fixed tuning budget, and the same metrics.
3. Measure what we gain when we add extra signals step by step: **text** $\rightarrow$ **+context** $\rightarrow$ **+image** $\rightarrow$ **+video keyframes**. Media experiments run on the linked subset and we report its split statistics.

4. Build and test a fusion model that still works when some modalities are missing and when classes are imbalanced, while staying within Google Colab or one consumer GPU such as an RTX 3070 or RTX 4060.
5. Report results in a transparent way, including ablation tables, efficiency notes, calibration checks, and paired tests for the most important comparisons.

Research Questions

1. Under one controlled protocol, how do classic ML baselines compare with transformer baselines? How much of the gain comes from adding context and media signals, instead of only upgrading the text encoder?
2. On the linked subset with media, do images and video keyframes raise recall for harmful content and improve macro-level metrics such as macro-F1 and AUPRC, compared with text-only and text+context models?
3. How well do models transfer across platform subsets? Do simple robustness steps, such as imbalance-aware training and a small-budget domain-adaptive pretraining step, help when writing style and topics shift?

Related Work

Surveys of cyberbullying detection show that there is already a large body of work in this area, but evaluation methods differ enough between studies to make direct comparison difficult; reproducibility and data scarcity are still common problems [4.][5.]. Pretrained transformers—BERT [1.] and RoBERTa [9.] in particular—have become the standard text encoders for classification tasks and provide strong baselines for transfer learning; however, higher accuracy on benchmarks does not always mean better robustness when the data distribution changes [13.]. Some recent work has also tested instruction-tuned large language models for hate speech and cyberbullying detection, with

mixed results: zero-shot performance can be reasonable on well-formed text, but often drops on informal or code-switched language, which is common in civic platforms [14].

However, it is important to note that multi-modal integration does not automatically guarantee performance gains. In many datasets, text alone explains a large share of the labels, and label noise can mask small improvements. Visual features may also encourage shortcut learning, for example a meme template that correlates with a class in one subset but not in another. Therefore, we treat the contribution of visual modalities as an empirical question, and we report both improvements and failure modes through systematic ablations and error analysis.

Domain-adaptive pretraining continues masked language model training on in-domain text before task fine-tuning. It is a practical approach that can reduce vocabulary and style mismatch [6.]. For class imbalance, focal-style loss re-weighting has been shown to make models pay more attention to harder, under-represented examples [8.]. Finally, significance testing is recommended when comparing models on NLP benchmarks, since differences in macro-level metrics are often smaller than they look [3.]; we adopt this practice in our evaluation plan.

Proposed Methodology

Dataset and Data Preparation

We will work with a multi-platform cyberbullying dataset with user sessions and post or comment text, plus a linked subset of media assets [11.]. The media inventory in the provided folders includes **2,263 images** in .jpg format and **983 short videos** in .mp4 format. Most media comes from Instagram images and Vine short videos.

Text and session or context files are stored as platform-specific user or session records, for example “common users” and “normal users”. Labelled subsets are provided for supervised learning.

Target labels. To keep comparisons clean, we mainly report a **binary** setting: harmful or cyberbullying versus non-harmful. We will document the mapping from original

labels to this binary label. If a subset supports finer labels, such as role, type, or target, we treat that as a secondary experiment and report it separately.

Linking text and media. When post or session identifiers exist, we link each text record to its image or video by matching shared IDs that appear in metadata files and media filenames. If linkage is incomplete, we still run the text+context pipeline for all records, and we run image and video experiments on the linked subset only. We report the linked subset size and its split statistics so readers can interpret results fairly.

Keyframe extraction for the video MVP. We keep video modelling simple. For each Vine video, we sample a small number of frames, starting with $K = 8$ or 16, spread uniformly across the video. We encode each frame independently and then pool the frame vectors with mean or max pooling to get one video representation.

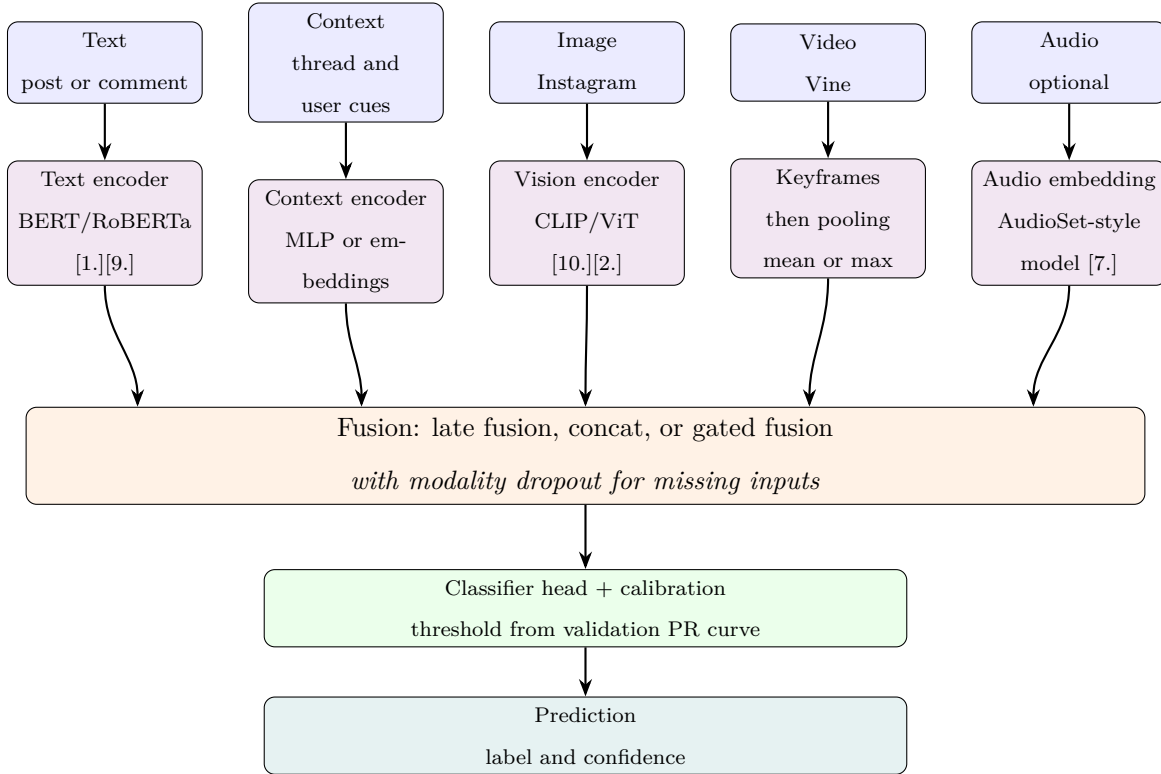
Audio signals. Audio is optional in this project. Short social videos do not always have a clean audio track, and extraction quality can vary. We will only use audio features if we can reliably extract audio from a reasonable portion of videos, and we will report the coverage. If audio coverage is low, we keep audio as future work.

Annotation quality check. Cyberbullying labels can be subjective. We will run a small quality audit by double-checking a random sample of 200–300 items. Two team members will follow a short guideline, and we will report inter-annotator agreement using Cohen’s κ plus a brief note on common disagreements.

Splits and leakage control. We use stratified train, validation, and test splits with a 70/10/20 ratio and a fixed random seed. When session-level identifiers exist, splitting is done at the session or user level to avoid near-duplicate leakage across splits.

System Overview and Model Flow: Figure 1

Figure 1 shows the end-to-end pipeline. All large encoders are used as frozen feature extractors; only small fusion and classification layers are trained in our core experiments.

**Figure 1**

Multi-modal pipeline. Text and context are the core input. Images and video keyframes are added as frozen vision features on the linked subset. Audio is optional and only used when we can extract it reliably.

Baseline Models

We benchmark representative baselines under one consistent protocol:

1. **Classic ML:** Logistic Regression and Linear SVM on TF-IDF with word and character n-grams.
2. **Transformer text:** Fine-tuned BERT and RoBERTa with a standard classification head.
3. **Text + context fusion:** concatenate a text representation, either TF-IDF or a pooled transformer embedding, with engineered context features, then train an MLP.

4. **Text + vision fusion:** add frozen vision embeddings from CLIP or ViT and compare late fusion of probabilities with feature concatenation.

Our Model and Proposed Improvements

Our proposed model keeps architecture simple but targets practical deployment issues:

- a) **Robust fusion:** gated fusion with **modality dropout**. We randomly hide one modality during training so the model stays stable when images or videos are missing at test time.
- b) **Imbalance-aware learning:** class-weighted loss and threshold selection on the validation precision-recall curve. We track harmful-class recall and macro-F1 [8].
- c) **Efficiency-first training:** freeze large encoders and cache embeddings; train only small fusion/head layers.

As an optional extension, we may try small-budget continued pretraining for the text encoder, often called DAPT [6.].

Training Protocol and Compute Budget

All experiments will run on **Google Colab** or on **one consumer GPU**, typically an RTX 3070 or an RTX 4060. We keep the training plan simple and realistic for a student team.

For **transformer text baselines**, we fine-tune BERT or RoBERTa with a small, fixed budget and early stopping. For **multi-modal fusion**, we freeze the large encoders and cache their embeddings to disk, so training reduces to a small fusion layer and classification head.

To stay within memory limits, we use mixed precision and small batch sizes. If needed, we use gradient accumulation to reach an effective batch size without exceeding VRAM. Typical settings are batch sizes between 8 and 32, depending on text length, and

we target 8–12GB VRAM usage on RTX 3070 or RTX 4060 class GPUs. We also cap the hyperparameter search with a small grid over learning rate, batch size, and fusion type. For feasibility, we report training time and peak GPU memory for the main runs.

At inference time, we keep the user-facing path responsive under the same compute limits. Text and images can be processed synchronously in seconds, while video keyframe extraction and scoring can be routed to an asynchronous job queue to avoid timeouts and smooth GPU usage.

Evaluation Plan

Primary metrics. For all models we report macro-F1, per-class precision and recall, and AUPRC [4.]. Because harmful content is rare, we pay special attention to harmful-class recall and to macro-level metrics.

Threshold selection and operating points. We select thresholds on the validation set using the precision–recall curve. Instead of one fixed cut-off, we report a few realistic operating points. For a citizen-facing self-check we aim for higher precision to avoid too many false alarms. For a moderator review queue we can prioritise recall, since a human confirms the final decision.

Ablations. We run a step-by-step comparison: text-only, text plus context, then adding images, then adding video keyframes on the linked subset. We also compare fusion choices, including late fusion, feature concatenation, and gated fusion.

Extra checks. We include calibration checks, such as ECE and reliability diagrams, so confidence scores are meaningful for threshold selection. For key comparisons we run paired significance tests [3.]. We also provide a short error analysis, focusing on common failure cases such as sarcasm, context-dependent targeting, meme-like images, and missing-modality inputs.

Project Timeline and Work Allocation

The project follows a 15-week schedule starting from January 9th, 2026, aligned with the course requirements. The detailed phases and deliverables are outlined in Table 1

and visualized in Figure 2.

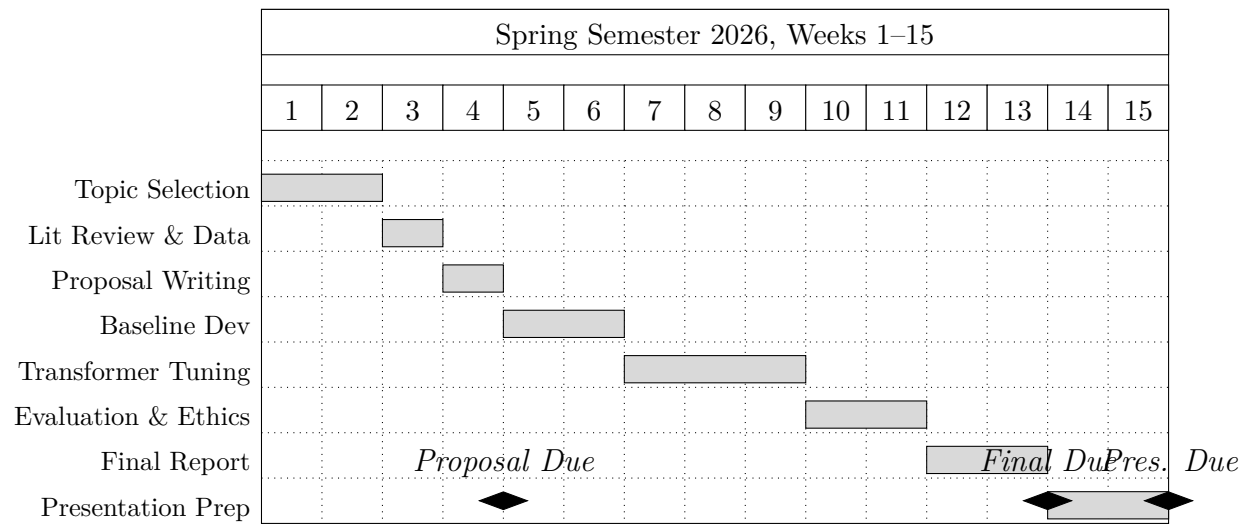


Figure 2
Gantt chart of the project lifecycle, from initialization to final presentation.

Table 1*Project Schedule (Jan 9 – May 3, 2026)*

Week	Date Range	Key Tasks	Deliverables
1–2	Jan 9 – Jan 26	Team formation, topic confirmation, and scope definition.	Topic Confirmation Form, Work Division Plan.
3–5	Jan 27 – Feb 16	Literature review; dataset acquisition and inventory; data cleaning; draft labeling guideline and pilot labeling.	Lit Review Notes, Data Inventory, Labeling Guideline v1.
6–8	Feb 17 – Mar 8	Proposal drafting and revision; finalize methodology, ethics, and evaluation plan; internal peer review and final edits.	Final Proposal, 30% of grade.
9–10	Mar 9 – Mar 22	Feature engineering and baselines; train classic models (Logistic Regression and SVM); set up experiment tracking and reporting.	Baseline Model Files, Training Log/Report.
11–12	Mar 23 – Apr 5	Fine-tune transformer baselines (BERT and RoBERTa); run initial multi-modal fusion on the linked-media subset; small-scale hyperparameter tuning on GPU.	Transformer Models, Fusion Checkpoints, Interim Comparison Table.
13	Apr 6 – Apr 12	Full evaluation and ablations; error analysis; ethics/compliance check; final report writing and polishing; Turnitin similarity check.	Final Project Report, 50% of grade.
14–15	Apr 13 – May 3	Prepare presentation and rehearsal; finalize packaging, documentation, and archiving.	Presentation PPT, Archived Project Assets, 20% of grade.

Work Allocation

This project is built by a group of five students from the MSc in Smart City Technology and Applications programme, also known as MScSCTA. We split work by technical component, but we still cross-check each other’s code and writing. In summary, while each member leads a specific technical component, the entire team participates in code review and final manuscript preparation.

- **Data preprocessing and baselines – ZHU:** Builds the data workflow, including deduplication, normalisation, stratified splitting, and label mapping. Implements the classic ML baselines, TF-IDF with logistic regression and linear SVM, which set the first reference point for later experiments.
- **Transformer modelling and tuning – TIAN:** Fine-tunes BERT and RoBERTa under the shared protocol. Keeps training logs and notes so we can compare runs fairly across model families.
- **Context features and data linking – GUO:** Extracts non-text cues such as timestamp bins, reply or mention statistics, and URL or media linkage identifiers when available. Makes sure these features are formatted consistently for fusion models.
- **Evaluation and robustness – WANG:** Implements the evaluation scripts, including macro-F1, AUPRC curves, confusion matrices, and harmful-class recall tracking. Runs robustness checks across platform subsets and reviews calibration outputs.
- **Reproducibility and versioning – WEN:** Maintains the shared codebase and file structure. Enforces fixed seeds and configuration logging so the main results can be re-run from documented steps.

Expected Outcomes

We expect to deliver:

1. A cleaned dataset split with documented label mapping, plus a reproducible training pipeline with scripts and configuration files.
2. A benchmark report comparing text-only baselines, classic ML and transformers, with multi-modal variants that add context, images, and video keyframes.
3. An efficient multi-modal fusion model using gated fusion and modality dropout, with an ablation study showing what each modality adds.
4. An evaluation summary including macro-F1/AUPRC, calibration and threshold selection, and a brief error analysis with representative false positives/negatives.
5. A short annotation-quality audit report. We report Cohen’s κ on a small double-checked sample and discuss common ambiguous cases.
6. A brief discussion of ethical limitations: known failure modes, for example higher false-positive rates for certain linguistic styles, risks of automated moderation for affected users, and recommendations for human-in-the-loop oversight.

Risks and Mitigation

- **Class imbalance and unstable minority metrics:** report macro-F1 and AUPRC; tune operating thresholds on validation data and discuss recall-oriented settings for moderation.
- **Incomplete text–media linkage:** run text+context experiments on the full set; run vision/video experiments on the linked subset and report subset size and split statistics transparently.
- **Subjective labels:** conduct a small double-check sample and report Cohen’s κ ; include an “ambiguous” bucket during auditing to document edge cases.

- **Compute constraints and operational costs:** we run on Colab or one GPU in the RTX 3070 or RTX 4060 range. We freeze encoders, cache embeddings, use mixed precision and gradient accumulation, and keep the hyperparameter search small. For deployment feasibility, we also propose simple cost and abuse controls, including content-hash caching to avoid re-evaluating identical viral items and user rate limiting at the API gateway. We report efficiency for the main runs.
- **Data privacy and ethical use:** All experiments use publicly released research datasets with documented terms of use. No personally identifiable information will be retained beyond what is provided in the original data files. Model outputs will not be used to profile or target individuals; the system is intended to assist, not replace, human moderation judgement.

Future Work and Stretch Goals

We prioritise a core system that is reproducible and feasible within the course timeframe. We also outline several stretch goals that would move the prototype closer to end-to-end multi-modal moderation. We acknowledge that these extensions introduce architectural complexity and may yield diminishing returns, particularly when dataset labels are noisy.

- **Adaptive frame selection for video:** uniform keyframe sampling is a crude proxy for full-video understanding. A compute-efficient upgrade is to select frames adaptively on a per-video basis, processing more frames only when the model remains uncertain. For example, AdaFrame learns to decide which frame to observe next and when to stop, reducing redundant computation while preserving recognition quality [12.]. A related direction is shot or scene segmentation followed by temporal modelling over segments, which can capture timing but typically requires cleaner labels and higher compute.
- **Audio cues when coverage is good:** if we can reliably extract audio, we can add automatic speech recognition transcripts, prosody features, and audio event embeddings.

These cues may help when harassment is spoken, implied by tone, or carried by background sounds.

- **Cross-modal attention:** instead of simple feature fusion, we can use cross-modal attention to align words with frames or audio segments. This is more flexible, but it is also more expensive and more sensitive to data size.

Conclusion

This proposal sets up a cyberbullying detection study for smart-city style digital platforms. Our objective is to run a controlled comparison under one shared protocol, and to report results in a form that can support practical moderation decisions. We will evaluate classic ML baselines, transformer text models, and fusion models that add thread and user context plus media features from images and video keyframes. We keep the compute plan realistic for a student team by freezing large encoders, caching embeddings, and training only small fusion and classification layers.

We also do not assume that multi-modal is always better. Text often carries most of the signal, and extra modalities can introduce noise or shortcuts. In our view, images and video frames are most likely to help on posts where the text is vague, sarcastic, or relies on memes. If the visual features do not help on some subsets, we will report that clearly and explain it with ablations and error analysis.

Finally, we frame the system around real deployment roles in a city platform. Citizens may want a careful self-check, content posters may benefit from feedback before posting, and moderators need a review queue with clear priorities. This is why we focus on operating points and calibration, not only on accuracy. A model with honest confidence scores and a sensible threshold is usually more useful than a model with a slightly higher macro-F1 but unstable behaviour.

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